

CIRQL — Build Plan

"An ocean of circles that never ends — you light it with real taps, glowing Wonders, and the friends in your Cirql."

CIRQL is the flagship: a persistent, magical, social world of concentric island-rings. You start at **The Hearth** and sail outward forever. This plan is the working guide — checkoff lists mirror the Linear project "**CIRQL — Flagship World**" (team CHR). **Scope of this pass:** "through the arcade beat" — a shippable playable slice; deeper social depth is Phase 2 at the bottom.

Progress log

- **M1 · The World & Movement — DONE & LIVE** (commit [b664883](#)). Walkable Hearth island: cottage, Wonders monolith, Ferra (NPC), Cirql lantern ring, movement, minimap, glow, interact → dialog/toast.
- **M2 · Identity & Persistence — DONE & LIVE** (commit [b09c79c](#)). `/cirql` saves + resumes (ring/position/avatar/name) via the `cirql` `game_progress` key; first-run character creator (avatar moved into CIRQL) + "Look" re-edit; expanded avatar (hat styles, aura glow, companions, bigger palettes). Verified in-browser (0 errors, tsc clean).
- **Next up: M3 · Quest System (backbone).**

Locked decisions (owner, 2026-07-06 — "ALL IN")

- **The Wonders** = the 50 arcade cabinets folded IN-WORLD as enchanted monoliths you *attune* to (press E → the cabinet opens as an in-world overlay; you never leave CIRQL). Found on the first quest, which is the movement tutorial.
- **One currency: SPARKS.** All rewards come out of the arcade's own system and unify under sparks. **Earn** ~a spark a play + a small capped performance bonus (and, Phase 2, real-world taps). **Spend** in an in-CIRQL reward catalog (avatar cosmetics, Hearth décor, perks). Sparks are the games *and* in-platform reward.
- **Avatar lives in CIRQL.** The chooser moves out of the arcade into a **first-run character-creation** step (sign-up / first play), with **lots more fun choices** — auras/glows, patterns, hats/crowns, companions — despite the circular art. Free at creation + spark-unlockable extras.
- **Your Cirql** = your friends, shown as **lanterns lighting your Hearth** (dim → beacon). Grown by **meeting people in-world** ("share a light", a chat-free beacon) **and inviting real friends** (rewarded-referral, reskinned as light).
- **Chat is channelled** so the feed never jams: **Global** (ambient) · **Party/Group** (campaigns & friends) · **Direct Messages**. Needs the live presence layer, so it becomes milestone **M8**.
- **Look:** supersampled 16-bit RetroEngine + the neon glow-up kit — "modern-but-16-bit". Approved 5-frame lookbook + playable in-game mockup.

Completed — reusable foundations (M0)

- ✓ `/town` websocket prototype — presence + move + chat proven (CHR-211)
- ✓ RetroEngine, supersampled (SS=1.5) (CHR-212)
- ✓ 50 arcade cabinets + registry + arcade-shell — become the Wonders (CHR-213)
- ✓ Avatars + shared joystick + `game_progress` save table (CHR-214)

M1 · The World & Movement — DONE

- ✓ `/cirql` route + lazy page scaffold (CHR-215)
- ✓ World engine: canvas render loop + camera (CHR-216)
- ✓ Concentric-ring data model + ring configs (CHR-217)
- ✓ The Hearth (ring 0) authored map + props (CHR-218)
- ✓ Player movement: WASD + click + joystick + edge collision (CHR-219)
- ✓ Minimap: concentric rings + fog + player dot (CHR-220)
- ✓ Ambient neon polish (glow-up kit) (CHR-221)

M2 · Identity & Persistence — DONE

- ✓ Cirql state model + save/load endpoints (`cirql` key) (CHR-222)
- ✓ Load-on-enter + autosave + resume (CHR-223)
- ✓ Character creation on first entry — move avatar chooser into CIRQL (CHR-242)
- ✓ Expand avatar customization — more choices, fun, circle-friendly (CHR-243)

M3 • Quest System (reusable backbone)

- Schema/migration: quests, objectives, player progress (CHR-224)
- Server: quest registry + progress API (CHR-225)
- Client: quest log + objective tracker HUD (CHR-226)
- Client: NPC quest-givers + waypoint markers (CHR-227)

M4 • First Quest Chain (onboarding = movement tutorial)

- Quest 1 — "Find Your Feet" (teaches movement) (CHR-228)
- Quest 2 — "The Lantern Path" (teaches interact) (CHR-229)
- Quest 3 — "The Wonders Door" (find + enter the arcade) (CHR-230)
- First-run onboarding scripting hook (CHR-231)

M5 • The Wonders (in-world arcade)

- Wonders interior + cabinet props from registry (all 50) (CHR-232)
- Press-E launches cabinet as in-world overlay (embed) (CHR-233)
- Return-to-world flow (no hardcoded `/arcade` back) (CHR-234)
- Score → sparks reward bridge (CHR-235)

M6 • Sparks Economy & Your Cirql

- Sparks currency + World Energy meter (CHR-236)
- Your Cirql: friends as lanterns around the Hearth (CHR-237)
- Invite-a-friend hook + "share a light" beacon stub (CHR-238)
- Earn sparks from play (spark-a-play + capped performance bonus) (CHR-244)
- Unify rewards → sparks (retire the arcade's separate reward system) (CHR-245)
- Spend sparks: in-CIRQL reward catalog (cosmetics / décor / perks) (CHR-246)

M7 • Playable Slice Hardening

- Full-loop smoke test (CHR-239)
- tsc clean + mobile pass + perf (CHR-240)
- Beta-test checklist entry + deploy (CHR-241)

M8 • Social — presence & chat channels

- Live presence in-world (`/ws/cirql` , room per ring) (CHR-247)
- Chat channel model: Global / Party / DM (CHR-248)
- Channel switcher UI + safety/moderation (CHR-249)

Architecture notes (as built in M1)

- `client/src/game/cirql-world.ts` — declarative rings/props. **Add a ring = add a config here.** Ring 0 = The Hearth; outer rings stubbed (minimap fog) until Phase 2.
- `client/src/game/cirql-world-engine.ts` — `CirqlWorldEngine` extends `RetroEngine` . Host hooks for later milestones: `onInteract(kind,prop)` , `onLocalMove(ring,x,y)` (autosave), `getState()/applyState()` (persistence), `setStats()` (sparks / your-Cirql HUD), `toast()` . Interact key = **E** (and Space).
- `client/src/pages/cirql.tsx` — host page + joystick + E/run controls. In DEV the engine is on `window.__cirql` .

Gotchas to respect

- **DB migrations:** `db:migrate` is broken on the shared Neon DB; `db:push` truncates. Apply additive DDL via a guarded script (quest tables, M3).
- **Dev server:** `tsx` does not hot-reload server code — restart after server edits (kill the port-5000 PID if stuck). Client hot-reloads via Vite.
- `apiRequest` returns a raw `Response` — always `.json()` it.

- **Cabinet back-buttons hardcode** `/arcade` — override when a Wonder launches inside CIRQL (M5).
- **Don't shadow RetroEngine base methods/fields** (`px` , `ring` , `last`) — world fields are `posX/posY/curRing` .
- **Deploy:** push to `chr-5-7-phase0-run-locally` → Render auto-deploys (~2–4 min); confirm the new `/assets/*.js` bundle hash.

Phase 2 — deferred (after the arcade beat)

- Real-world tap → sparks bridge (the moat: a café tap widens the shared sea).
- Friends/parties depth — party up, sail together, co-op ring unlocks.
- Trading + inventories (both-confirm, anti-scam).
- More rings/biomes (Whisperwood, Glimmer Bazaar, outward forever) + docks/portals.
- Depth — quests beyond onboarding, light combat/monsters, seasonal drops.